



Home cinema A/V systems

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These days there are more possibilities for 'the TV' than just being a box in the corner of the room. It makes sense to install an integrated system, where the screen is at the hub of work and entertainment activities. Although this concept was first proposed several years ago it is only now gradually becoming economically and technically possible. Of course this does require more commitment, work and knowledge to set up than a simple TV. One thing definitely to bear in mind is that standards and features for digital audio and video are still changing so fast that it is difficult to make equipment choices that provide future-proofing. In this article we will take a look at component choices and how to best hook them together.

Choosing components

A home Audio/Visual system can be quite complex so it's best to plan in advance before purchasing any components. You will need to decide on what you require your home system to do and on how the available space and budget may restrict your choices. It may seem attractive to integrate equipment you already own, but because of the new features required by digital audio, digital video and HD video this may not be practical.

A system may be composed of the following components: a display, a media player and/or a gaming console, a set top box, an audio amplifier, loudspeakers (from two to eight), a hard disk or DVD recorder, a PC and remote controls, plus possibly an aerial, cable feed or satellite dish. You might also want to include a tuner for radio reception.

The display

For displays there is a choice to be made between a computer monitor, a television or a projector. Computer LCD monitors are available with up to a 50-inch diagonal screen size although these are expensive at nearly Rs. 3 lakh. However, you can, for example, get a ViewSonic 22-inch with an HDMI connector and HDCP support for around Rs. 15,000. The advantages of computer monitors are that in the smaller sizes they are relatively cheap and provide high resolutions. They are best suited to their primary purpose as computer displays with a secondary use as entertainment displays. However, for use in a home A/V centre it is important to buy a monitor with an HDMI input and HDCP support. There are still many new computer monitors for sale that do not have these features.

TVs have the advantage that they can be used stand-alone, are available with up to a 52-inch diagonal display and the current flat panel TVs usually have quite a range of input and output connectors. Again an HDMI input and HDCP support is essential; many of the earlier high resolution flat panel TVs do not have these features.

The big advantages of projectors are that they can be installed unobtrusively, they don't take up much space and they can provide large displays. Unfortunately, there are quite a few disadvantages: projectors are relatively expensive, particularly for higher resolutions and the running costs due to power consumption (350W for a 1200 lumen projector) and the cost of replacement lamps, are high. You can expect to pay between Rs. 60,000 to Rs. 130,000 for a 1920x1080 resolution. Replacement lamps can cost Rs. 18,000 to Rs. 25,000 and you should budget for

at least one lamp change during the life of a projector. Ceiling mounting a projector can also be something of a chore.

The Audio Amplifier

Let's avoid all the audiophile arguments about audio quality – frequency response, distortion and so on – and assume that an amplifier bought from a well known brand name manufacturer is going to be of a good enough quality and we'll concentrate on the required features.

The first thing to decide on is whether you might be satisfied with two channel stereo sound, or want to go for a surround set-up (with 5.1 or even 7.1). Most surround sound capable amplifiers have built-in decoding for both Dolby and DTS systems; best to avoid any that don't.

Modern loudspeakers are usually quite inefficient (efficiency is sacrificed in favour of flat frequency response). Many multi-channel amplifiers claim 100 or 200 W per channel although they don't specify exactly what is meant by this. 50 W r.m.s per channel will be more than enough in most domestic settings.

Remote control of volume and input selection are desirable features, but it's unlikely you will find a mainstream audio amplifier that doesn't have these.

Choose an amplifier with at least two HDMI inputs and two HDMI outputs; most amplifiers with these will also have a selection of phono connectors for other audio inputs and may also have digital optical inputs and outputs.

Loudspeakers – 5.1 and beyond

Although you can use the amplifiers and loudspeakers built in to a computer display or a television, these are normally limited to left and right stereo, are usually small and of limited quality. An external amplifier can drive multiple loudspeakers for a surround sound setup.

With stereo sound where two loudspeakers are placed one to the left and one to the right in front of the listener, the illusion of a sound coming from any point between the two loudspeakers can be generated by driving both loudspeakers with the same signal, but at different volumes. When the volumes are equal the sound source appears to be in the centre, if the left portion of the signal is louder the sound source appears to move to the left, if the right signal is louder the source appears to move to the right. In surround sound this principle is extended by adding more loudspeakers placed around the listening position and the greater the number of loudspeakers used the more accurate is the illusion. Movie theatre installations have used as many as 24 loudspeakers.

Home installations normally use five small 'satellite' loudspeakers plus one bass-reflex sub-woofer and are referred to as 5.1 (five smaller loudspeakers plus the single bass woofer). This arrangement is based on the principle that the sense of direction of a sound diminishes as frequency falls, so the bass frequencies can be provided by a single loudspeaker (note the difference between this and the movie theatre use of the sub-woofer only for extreme low frequency effects). Conveniently, it's the bass frequencies that require the largest, most powerful loudspeaker, while the middle to upper frequencies require less power and can be generated by much smaller loudspeakers. The two front loudspeakers provide the left and right sound stage for sounds in front of the head, while the two rear loudspeakers expand the